Performance test and simulation of a reciprocating engine for long endurance miniature unmanned aerial vehicles

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Abstract: Development of an engine with good fuel economy is very important for successful implementation of long endurance miniature UAVs (unmanned aerial vehicles). In the study, a four-stroke glow-plug engine was modified to a gasoline-fuelled spark ignition engine. Engine tests measuring performance and friction losses were conducted to tune a simulation program for performance prediction. It has been found that excessive friction losses are caused by insufficient lubrication at high speeds. The simulation program predicts that engine power and fuel economy become worse with high altitude, due to an increasing portion of friction losses. The simulation results suggest quantitative guidelines for further development of a practical engine.

Keywords: UAV, engine, friction, simulation, combustion, t.f.m.e.p., altitude, glow-plug, valve, gas exchange

1 INTRODUCTION

It has been proved that a miniature unmanned aerial vehicle (UAV) within 15 kg of its maximum take-off weight is technologically feasible for long endurance, which is termed as about 24 hours, surveillance, monitoring, and photographing by virtue of the development of a light and compact mission apparatus [**1**]. The UAV (named 'Laima') prepared by Aerosonde Company and Insitugroup Company has succeeded in crossing the Atlantic with a flight of 3270 km in a time of 26 h 45 min. The flight was done on a gallon and a half of fuel (5.7 litres) [**2**].

A technical obstacle to commercializing such UAVs, however, is the availability of an engine developed solely for long endurance. Even the engine employed in Laima was a considerably modified version from a two-stroke 'off-the-shelf' engine of the 3W Company in Germany for a 100 octane low lead fuel/oil mixture. An optimal requirement for realizing such long endurance is the adoption of a four-stroke engine and gasoline as fuel. Such an engine is under development [**3**].

Therefore, no test experiences are available in the literature for four-stroke gasoline engines with the displacement of around 20 to 30 cm^3 for mini-UAVs. The purpose of the study is to investigate performance characteristics from tests of such an engine and develop a performance simulation program for engine development and for prediction of engine performance at high altitudes. As a test engine, a four-stroke glow-plug engine was modified to a gasoline version.

Due to lack of published test experiences, many trials and errors were encountered. An empirical relation to predict friction losses is suggested from the measurement of friction losses by engine motoring tests. A performance simulation program is proposed which is slightly modified from a well-proven automotive engine simulation program and performance test data. It was found that a most critical factor affecting engine performance is friction losses and that its effect on fuel economy and power becomes worse with altitude and engine speed.

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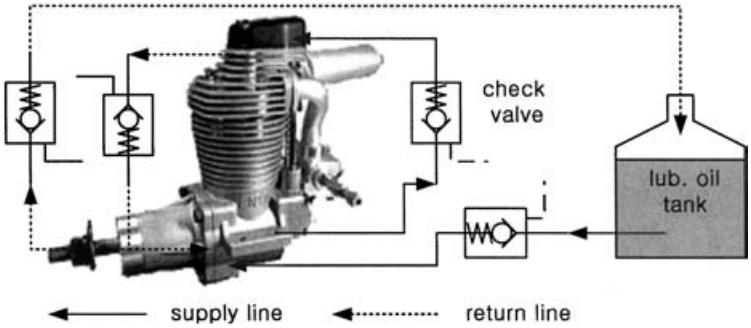

Fig. 1 Gasoline engine version of ENYA R155-4c with the lubrication system

2 TEST ENGINE AND EXPERIMENTAL APPARATUS

The base engine for performance simulation was Model R155-4c of the ENYA Company in Japan, which is a four-stroke glow-plug combustion engine. The engine was converted to a spark ignition version to use gasoline. To run on gasoline, the engine had to be equipped with a lubrication system as shown in Fig. 1. The lubrication was made possible by exploiting positive pulsating pressures in the crankcase with check valves. The test engine is shown in Fig. 1 and its major specifications are listed in Table 1.

Table 1 Specifications of ENYA R155-4C

Item	Measurement
Displacement volume (cm^3)	25.4
Bore (mm)	33.8
Stroke (mm)	28.3
Compression ratio	6.3
Intake valve diameter (mm)	14.3
Exhaust valve diameter (mm)	12.0
Intake valve lift (mm)	4.0
Exhaust valve lift (mm)	3.3

Engine power was measured with an air-cooled eddy current dynamometer of capacity 3.7 kW. The installation is shown in Fig. 2. The engine was operated with an in-house engine control unit (ECU) specially designed for the UAV. The air–fuel ratio was measured with a lambda sensor. Measurement of friction losses with engine speed is very important in order to tune a performance simulation program and understand frictional behaviour of the test engine, which is substantially different from a typical automotive engine in size and design. To measure the friction losses, a 3 kW a.c. servomotor was connected to the engine crankshaft via a 4 : 1 speed-up gear to motor the engine up to 12 000 r/min.

3 ENGINE SIMULATION MODEL

The engine simulation model is based on the program that has been gradually improved over years in the Sloan Automotive Engine Laboratory at MIT [**4**, **5**]. Full modelling of combustion in a three-dimensional numerical scheme is quite demanding in terms of computational load and modelling

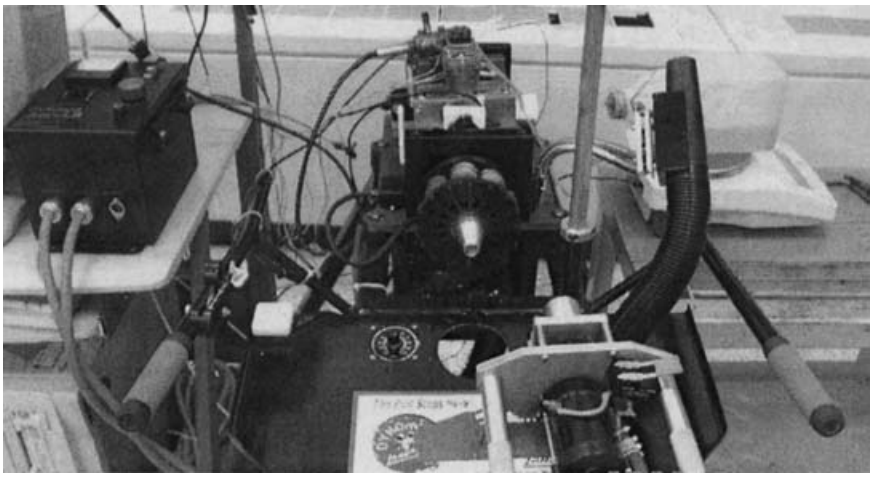

Fig. 2 Test set-up for the measurement of engine power

difficulties. Since the outcome expected from the simulation in this study is a reasonable prediction of measured engine power, combustion has been simplified as a quasi-dimensional model. In the model, a flame front propagates as a spherical surface by which the combustion chamber volume is divided into burnt and unburnt regions. The burnt region is modelled in a way that heat is transferred across the boundary layers in turbulent convection and the heat loss affects engine performance and burnt gas composition.

Although the program is capable of predicting flame propagation speeds from experimental relations for turbulent flame propagation, the prediction has been found to be impractically poor. Therefore the following Wiebe empirical relation between the mass fraction burnt and the engine crank angle has been applied [**6**] in the form

$$\chi_b = 1 - \exp\left[-a\left(\frac{\theta - \theta_0}{\Delta\theta}\right)^{m+1} \right] \tag{1}$$

The total combustion duration $\Delta\theta$ and the data fitting constants a and m were set to 55 CA, 5.0 and 2.0 respectively by adopting values closest to experimental data.

The heat transfer through the combustion chamber walls has been modelled by the following non-dimensional relation regarding turbulent convective heat transfer in which the instantaneous heat transfer area wetted by the flame front is calculated at every crank angle

$$Nu = c\, Re^{0.8} Pr^{0.33} \tag{2}$$

The characteristic velocity V needed to calculate the Re number has been defined as follows in terms of the mean fluid velocity U, turbulent intensity u', and instantaneous piston speed V_p

$$V = \left[U^2 + u'^2 + \left(\frac{V_p}{2}\right)^2 \right]^{1/2} \tag{3}$$

where the turbulent intensity u' is calculated from the well-known κ–ε turbulence model. The κ–ε model is one of the most classical turbulence models. It relies on a two-equation model. The turbulence kinetic energy κ is computed from the transport equation; meanwhile the turbulent energy dispersion ε is given by an algebraic expression.

Modelling of the gas exchange process through intake and exhaust ports is needed to calculate volumetric efficiency. The process is described using the following well-known orifice model for the subsonic region of a compressible fluid

$$\dot{m} = c_d A \frac{p_0}{RT_0}\sqrt{\gamma R T_0}\left\{ \frac{2}{\gamma - 1}\left[\left(\frac{p_s}{p_0}\right)^{2/\gamma} - \left(\frac{p_s}{p_0}\right)^{(\gamma+1)/\gamma} \right] \right\}^{1/2} \tag{4}$$

and the sonic region of a compressible fluid

$$\dot{m} = c_d A \frac{p_0}{RT_0}\sqrt{\gamma R T_0}\left(\frac{2}{\gamma + 1}\right)^{(\gamma+1)/2(\gamma-1)} \tag{5}$$

The instantaneous valve throat area A was calculated from the information of the typical valve lift profiles of an automotive engine [**4**] and valve diameters and maximum valve lifts given in Table 1. The valve lift profiles were obtained by normalizing the typical automotive profiles with their maximum valve lifts and then dimensionalizing again the normalized one with the maximum valve lifts given in Table 1 by taking into account the observation that most cam profiles are generically similar in plug-assisted ignition engines. In addition, discharge flow coefficients with a valve lift are necessary to improve accuracy in calculating the instantaneous mass flow-rate [**7**]. They were determined from the dimensional analysis and the experimental database of discharge coefficients from the Ford Company [**4**].

Another important factor to improve simulation results is application of actual valve timing. The valve timing of the test engine is as follows: IVO/IVC/EVO/EVC (intake valve open/intake valve closed/exhaust valve open/exhaust valve closed) = 30° BTDC (before top dead centre)/70° ABDC (after bottom dead centre)/70° BBDC (before bottom dead centre)/30° ATDC (after top dead centre). Compared to the valve timing of a typical automotive engine (IVO/IVC/EVO/EVC = 13° BTDC/55° ABDC/57° BBDC/3° ATDC), it can be seen that the valve opening periods are longer than those of a typical automotive engine to secure the gas exchange process at high speeds.

4 ENGINE FRICTION LOSS

Since it is difficult to predict accurately friction losses, an engine test to measure friction losses must be conducted to secure the accuracy of the engine performance simulation program. It is especially the case for the 26 cm^3 test engine since there has been no report of friction loss measurements for this category of engine. Any empirical relation for automotive engines cannot be directly applied to the 26 cm^3 test engine because it is very simple in hardware design compared to automotive engines and very poor in lubricating performance. Thus it is

necessary to derive its empirical relation from engine motoring tests.

The following equation is an empirical relation representing engine friction loss as a function of engine speed N (in r/min)

$$\text{t.f.m.e.p.} = a + b\left(\frac{N}{1000}\right) + c\left(\frac{N}{1000}\right)^2 \quad \text{(kPa)} \qquad (6)$$

In the equation, the terms on the right-hand side represent boundary lubrication, hydrodynamic lubrication, and flow resistance of the intake air respectively. The resulting frictional loss obtained from experiments of the UAV test engine is shown in Fig. 3. The following empirical relation for a typical passenger car engine is also plotted in Fig. 3 for comparison [**8**]

$$\text{t.f.m.e.p.} = 68.6 + 19\left(\frac{N}{1000}\right) + 2.1\left(\frac{N}{1000}\right)^2 \quad \text{(kPa)} \qquad (7)$$

When it comes to the test engine, m.m.e.p. (motored mean effective pressure) with the throttle valve wide open was plotted instead of f.m.e.p. (frictional mean effective pressure). Although pumping loss is small at wide-open throttle conditions, p.m.e.p. (pumping mean effective pressure) has also to be taken into account to improve simulation accuracies. The plotted p.m.e.p. was calculated from the gas exchange process of the simulation program. The reasons why measured m.m.e.p. below 5000 r/min in Fig. 3 are scattered in a wide range are threefold. Firstly, torque variations resulting from a cyclic cylinder pressure variation become pronounced at low speeds associated with longer cyclic periods. Secondly, the small engine is not equipped with a flywheel that serves to dampen the cyclic torque variation. Lastly, the sampling interval of the datalogger employed in the experiments was 2 s. Those factors resulted in the capture of all possible instantaneous torque variations, as shown in Fig. 3. Ideally the measured torque should have been lowpass-filtered long enough so that the logger sees a cyclic average of the instantaneous torque. In other words, the time constant of the low-pass filter should have been set to about 2 s. Although the remedy has not been made, the scattered data in Fig. 3 is more meaningful than the low-pass filtered average, in that it makes it possible to appreciate the magnitude of torque variation with engine speed and that the data size is large enough to take correct averages.

One thing to note for the test engine is that it is not possible to curve-fit the measured data with a single second-order polynomial such as equation (6). The cause can be explained as follows. Friction loss of a general automobile engine may fit in a single empirical equation since the maximum speed of automobile engines is limited by around 6000 r/min. However, for remote-controlled (R/C) glow-plug engines whose rated speed is between 10 000 and

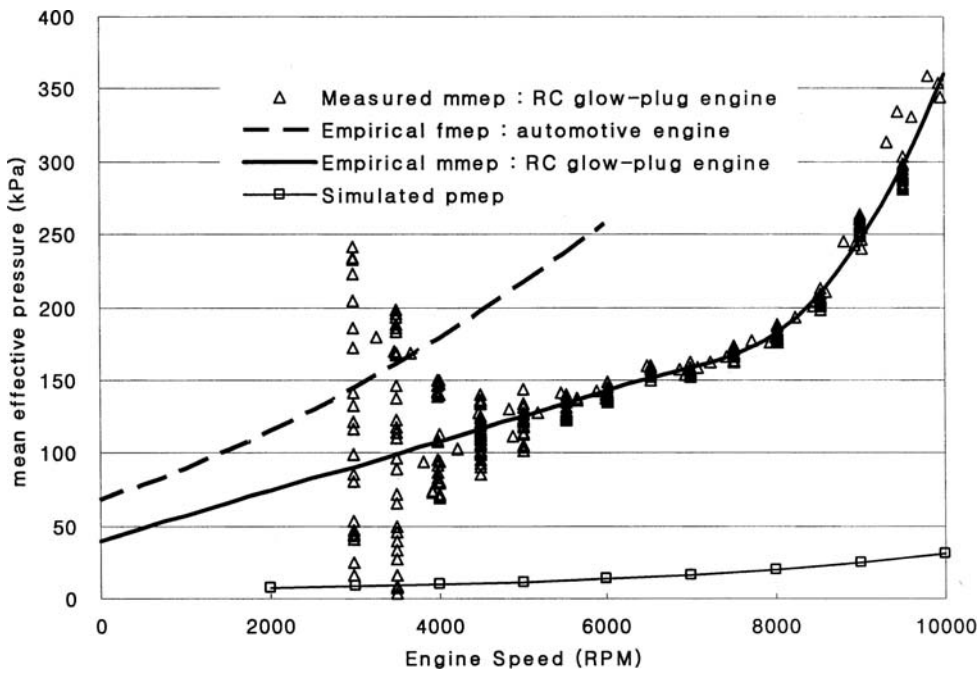

Fig. 3 Measured friction losses of the test engine in comparison with those of a typical automotive engine

12 000 r/min, frictional losses in the region above 8000 r/min are observed to increase so drastically that they cannot fit in a single empirical equation. Therefore the data up to 8000 r/min was curve-fitted with a first-order polynomial because the mean piston speed of the R/C engine was about one-third that of typical automotive engines. The slower piston speed means that frictional loss associated with hydrodynamic lubrication, which is proportional to engine speed, is more dominant than the loss caused by aerodynamic resistances for the speed range. However, for the range above 8000 r/min, the frictional mechanism changes drastically. The phenomenon was observed to be consistently repeatable in several test trials with and without a silencer. The phenomenon observed in Fig. 3 cannot be explained only by the assumption that aerodynamic friction starts to increase rapidly at high engine speed because its effect is too gradual to fit all the data in a single second-order equation. Furthermore, simple calculation reveals that the engine speed for any possible intake chocking is far above tens of thousands r/min. This means that there must be a cause that cannot be explained in typical automotive engines.

The cause can be associated with the poor lubrication system. R/C engines have no lubrication pump so at high engine speed they may not be lubricated enough. The insufficient lubrication leads to a mixed form of boundary lubrication (caused by direct contact of two surfaces) and hydrodynamic lubrication. The mixed form of lubrication seems to play a role from about 8000 r/min where significant deviation from any form of a single second-order polynomial curve-fit starts to be pronounced. In short, hydrodynamic lubrication is dominant below 7500 r/min and the mixed lubrication due to insufficient oil supply becomes important above 7500 r/min. For smoother transition between the two lubrication regimes, the following two empirical equations had to be proposed

$$\text{m.m.e.p.} = 40 + 17\left(\frac{N}{1000}\right) \quad (\text{kPa}) \qquad (N < 7500\ \text{r/min}) \tag{8}$$

$$\text{m.m.e.p.} = 40 + 17\left(\frac{N}{1000}\right) + 24\left(\frac{N}{1000} - 7.5\right)^2 \quad (\text{kPa}) \qquad (N \geqslant 7500\ \text{r/min}) \tag{9}$$

The p.m.e.p. calculated from the simulation program is fitted with the following linear relation

$$\text{p.m.e.p.} = 2.7\left(\frac{N}{1000}\right) \quad (\text{kPa}) \tag{10}$$

Finally, f.m.e.p. is obtained from the following relation

$$\text{f.m.e.p.} = \text{m.m.e.p.} - \text{p.m.e.p.} \tag{11}$$

5 SIMULATION RESULTS

Figures 4 and 5 show the comparison of torque and power between the test results and the simulation

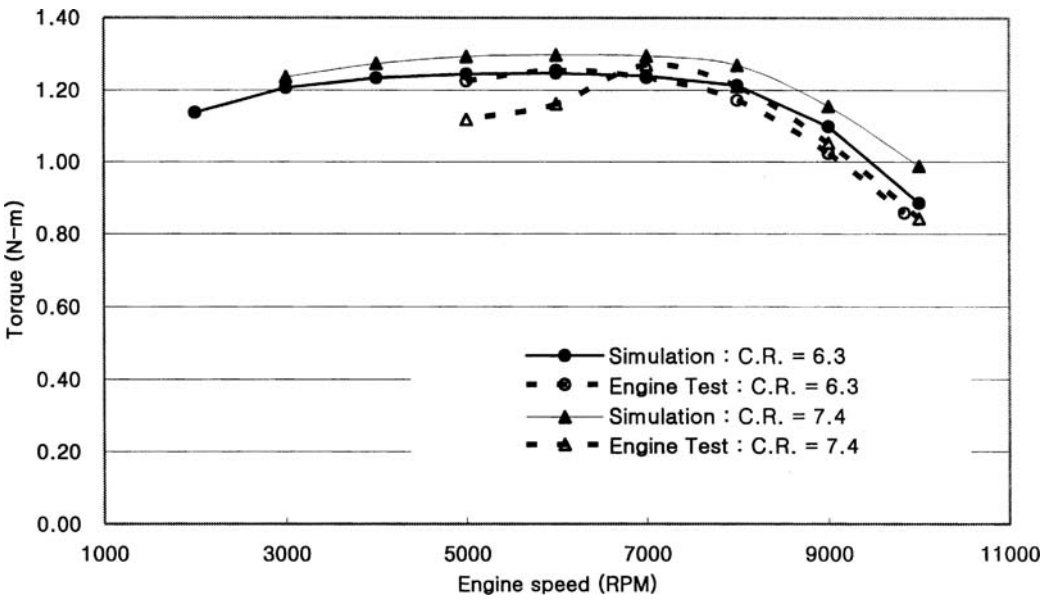

Fig. 4 Comparison of engine torque at wide-open throttle

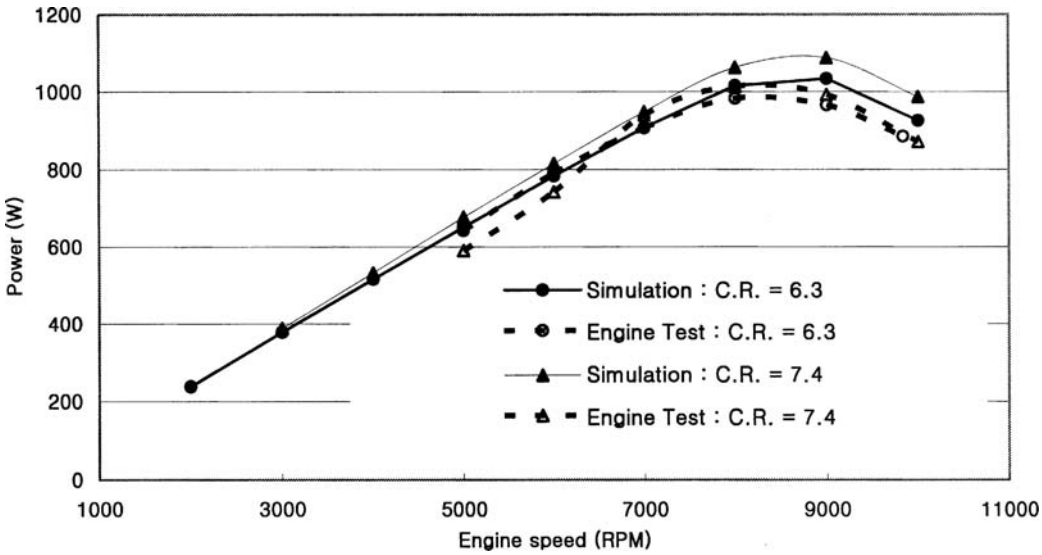

Fig. 5 Comparison of engine power at wide-open throttle

results with respect to compression ratio respectively. It is seen that they are in good agreement. In the comparison, the simulation results pronounce the effect of the compression ratio, whereas it is obscure in the measured torque and power. The smaller increase in torque and power in the experiments is ascribed to the simple design of the R/C engine in which a single compression ring does the job of gas sealing as well as oil scraping as compared with an automotive engine where two compression rings and an oil ring do the same job. The use of a single piston ring leads to excessive gas leakage and hence less torque and power increase than expected from the increase in the compression ratio.

Another phenomenon to note is the pronounced decrease in torque measurements for the compression ratio of 7.4 in the region below 7000 r/min. It was quite repeatable in similar experiments. It also seems to be associated with the use of a single piston, which makes the combustion chamber leaky. It also explains why the production engine has adopted the lower compression ratio of 6.3, which mitigates mixture leakage during the compression process.

The simulation predicts power output for the gasoline version converted from the original ENYA R155-4C glow-plug engine. In the simulation, cylinder wall temperatures were set to the ones measured at the base of air cooling fins. The ambient condition was set to the atmospheric temperature and pressure of the corresponding altitude. Engine friction was estimated from the empirical relations (8) to (10). The engine simulation model introduced in the previous section predicts i.m.e.p. (indicated mean effective pressure) only; b.m.e.p. (brake mean effective pressure) is the mean effective pressure on the piston obtained by subtracting f.m.e.p. and p.m.e.p. from i.m.e.p.

Figures 6 and 7 show the variation of b.m.e.p. and power with altitude. In general, it is known that i.m.e.p. at rated power is around 1100–1200 kPa for modern naturally aspirated gasoline engines [**9**]. According to the simulation result for the compression ratio of 6.3, the estimated i.m.e.p. is about 800 kPa over all the speed range. Therefore, it is clear that the drastic decrease in b.m.e.p. above 8000 r/min in Fig. 6 is due to excessive friction losses. According to Fig. 7, the rated power on the ground is about 1 kW at 9000 r/min. The power is comparable to that of the 24 cm^3 Aerosonde UAV engine that has produced 1 kW with aviation gasoline. Considering the fact that the test engine has a low compression ratio of 6.3 and its carburettor is not optimized, it is concluded that it produces reasonable power output in spite of its larger displacement volume of 25.4 cm^3. The power drop with altitude is quite noticeable in Fig. 7. This is because the intake air density decreases with altitude. Therefore UAVs should fly as low as possible to maintain a high power output.

Figure 8 shows the comparison of brake specific fuel consumption (b.s.f.c.) with tests and simulation results on the ground. The measured b.s.f.c. is higher than that of the simulation. Since comparable power between the test and the simulation in Fig. 5 means that thermal efficiencies are also comparable, the b.s.f.c. predictions should also be close to the measurements. The discrepancy in b.s.f.c. could be due to

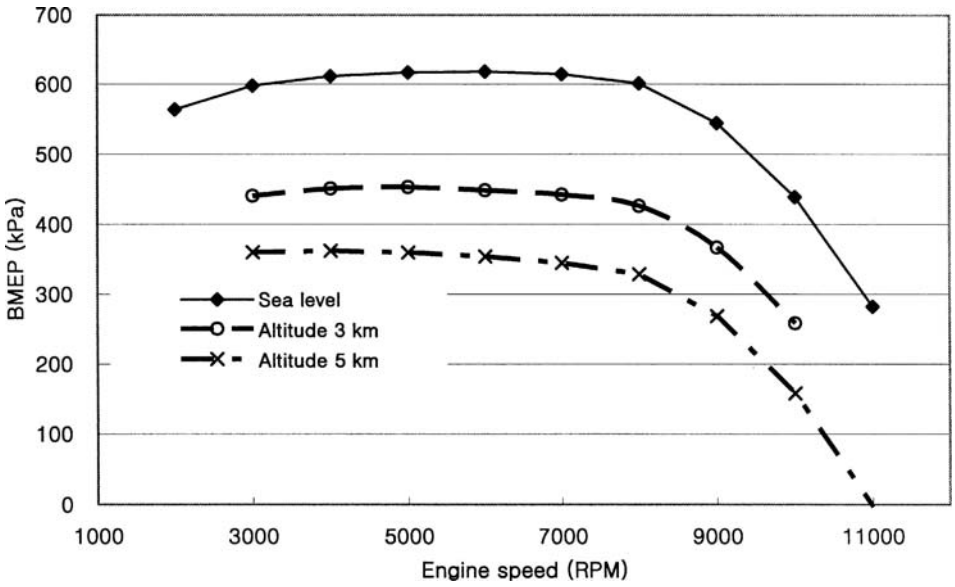

Fig. 6 Simulation result of b.m.e.p. with altitude

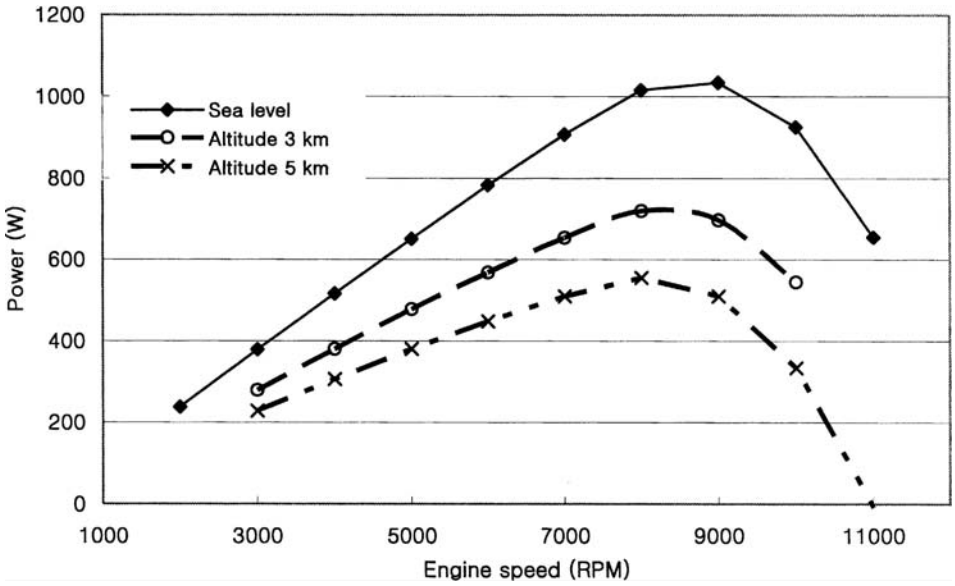

Fig. 7 Simulation result of engine power with altitude

an inaccurate air–fuel ratio read by a lambda sensor that is installed in the special exhaust reservoir. The reservoir, as big as the engine, is designed to allow lambda sensor and exhaust gas temperature measurements to be made. The reservoir was securely bolted on to the exhaust port. However, the heavy reservoir was subject to continuous engine vibration which caused loosening of the bolts on the exhaust flange and hence air leakage. This led to a lean shift in the air–fuel ratio. A relatively low exhaust gas temperature could also have contributed to the lean shift. The carburettor also gave only coarse control of the air–fuel ratio. Therefore, estimating b.s.f.c. based on the simulation could be more plausible.

Figure 9 shows the comparison of b.s.f.c. with altitude. It is seen that the fuel consumption increases with altitude. It can be explained as follows. Air density reduces with altitude, leading to a reduced mass of air and fuel; hence there is a reduction in power. On the other hand, friction losses are nearly constant with altitude since they are mainly a function of engine speed. Therefore, increased friction losses together with the reduced power output makes the specific fuel consumption get worse.

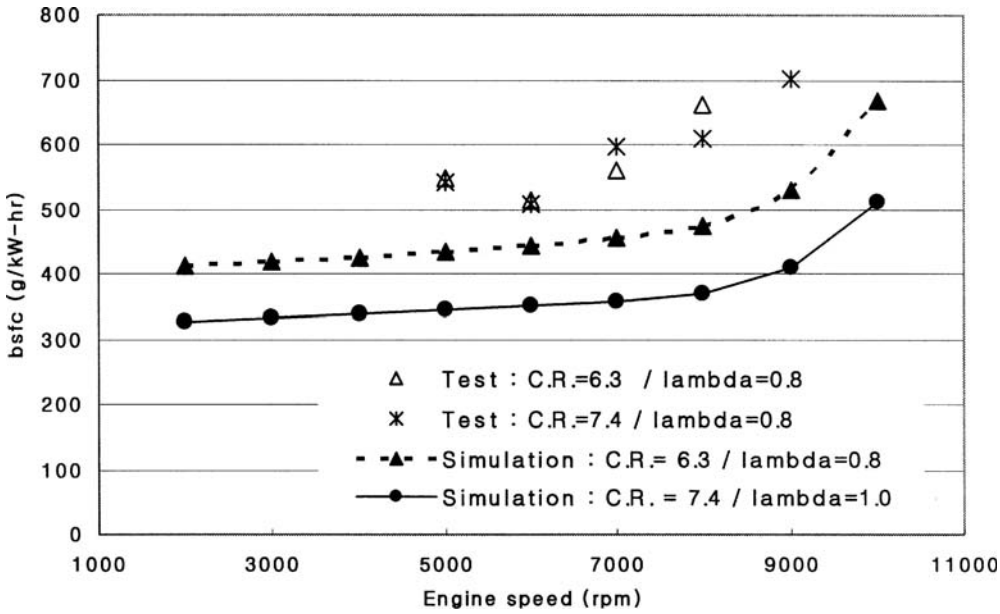

Fig. 8 Comparison of fuel consumption with engine speed for the test and simulation

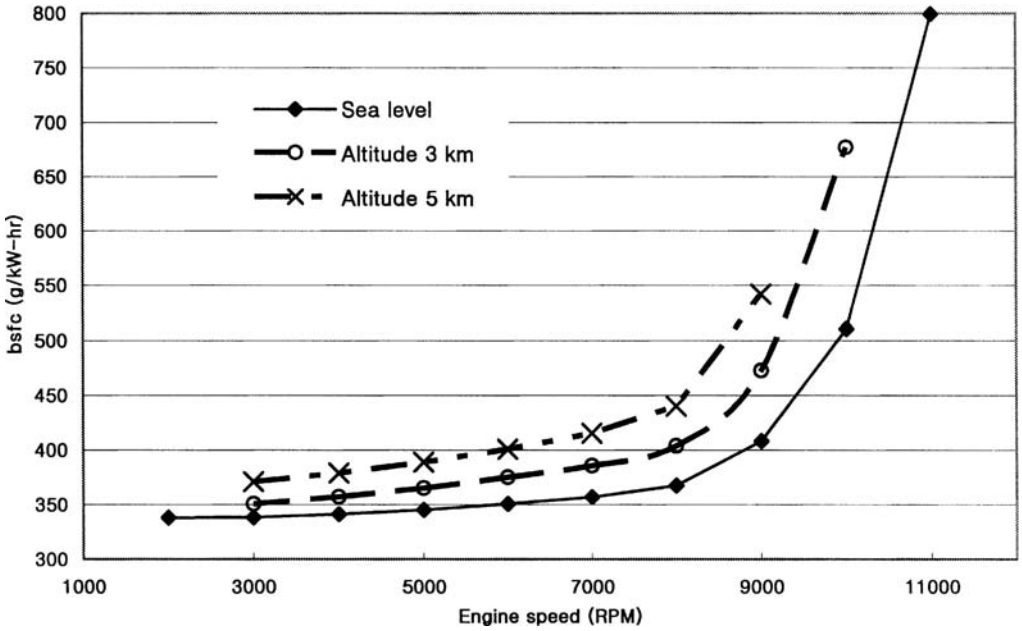

Fig. 9 Simulation result of fuel consumption with altitude

6 CONCLUSIONS

In order to develop an engine with high power and low fuel consumption, it is essential to predict quantitative engine performance at altitudes where a UAV cruises. The contribution of this study can be found in the following. The study has measured performance of an engine that belongs to a class of small four-stroke reciprocating engines whose performance measurements are not published in the literature. A performance simulation program has been developed by modifying a well-known automotive engine simulation program and by tuning simulation parameters with measured data of performance and friction losses. It is found that a significant power drop at high engine speed is caused by excessive friction losses and that the power loss is more pronounced at high altitude. Fuel consumption also increases with altitude and engine speed because of the increased proportion of friction losses. Therefore, it is recommended that the engine lubrication system should be improved in order to

achieve improvements in power and fuel economy. The results of the study can be applied to predict quantitative engine performance for engine development and finding optimal flight conditions of UAVs.

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REFERENCES

1 **McGeer, T.** and **Vagners, J.** Wide-scale use of long-range miniature aerosondes over the world's oceans. In Association for Unmanned Vehicle Systems International (AUVSI) 26th Annual Symposium Conference Proceedings, AUVSI, Baltimore, 1999.
2 **Finn, A., Brown, K.,** and **Lindsay, T.** Miniature UAVs and future electronic warfare. Land Warfare Conference, Brisbane, Australia, 22–24 October 2002.
3 http://www.insitugroup.com/pages/Technology/engine.html.
4 **Poulos, S. G.** The effect of chamber geometry on SI engine combustion rates—a modeling study. Master thesis, Department of Mechanical Engineering, MIT, Cambridge, Massachusetts, 1982.
5 **Heywood, J. B.** *Internal Combustion Engine Fundamentals*, 1988, pp. 389–390 (McGraw-Hill, New York).
6 **Heywood, J. B., Higgins, J. M., Watts, P. A.,** and **Tabaczynski, R. J.** Development and use of a cycle simulation to predict SI engine efficiency and NO_x emissions. SAE paper 790291, 1979.
7 **Woods, W. A.** and **Khan, S. R.** An experimental study of flow through poppet valves. *Proc. Instn Mech. Engrs*, 1965–1966, **180**(3N), 32–41.
8 **Labreuche, G.** *et al.* Total friction effective pressure and torque estimator. In The International Workshop MECA01, University of Salerno, Italy, 2001.
9 **Heywood, J. B.** *Internal Combustion Engine Fundamentals*, 1988, pp. 50–51 (McGraw-Hill, Maidenhead).

APPENDIX

Notation

A	flow cross-section (m^2)
C_d	discharge flow coefficient
$\dot{m}$	mass flowrate (kg/s)
N	engine speed (r/min)
Nu	Nusselt number
p	pressure (Pa)
Pr	Prandtl number
R	gas constant of a compressible fluid (J/kg K)
Re	Reynolds number
t.f.m.e.p.	total frictional mean effective pressure (kPa)
u'	turbulence intensity (m/s)
U	mean fluid velocity (m/s)
V	characteristic velocity (m/s)
V_p	instantaneous piston speed (m/s)
γ	specific heat ratio
θ	crank angle (CA)
θ_0	ignition timing (CA)
χ_b	mass fraction burnt

Subscripts

o	upstream stagnation
s	static at throat